Since AI 2025
Valmet challenge – detection of field devices

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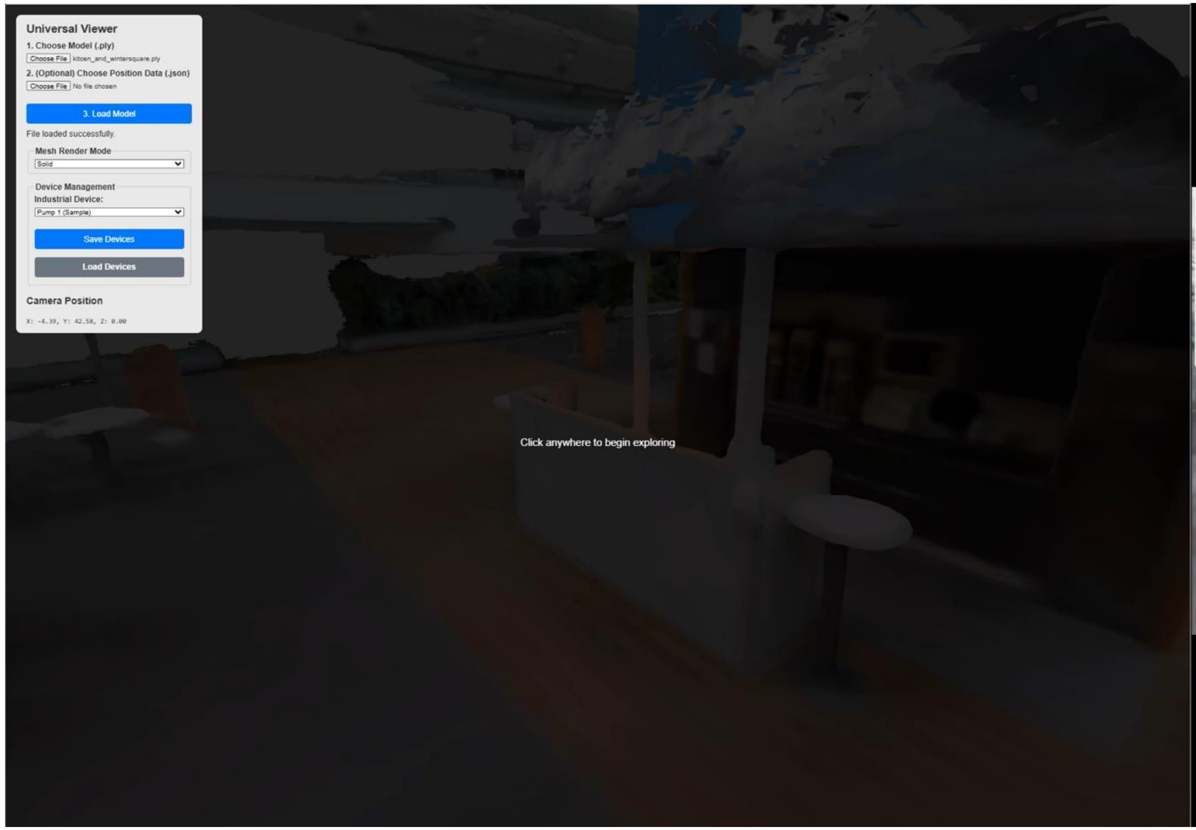
Agenda

- Traditional scene building: static model and content pre-programmed (like training environments)
  - Costs can be high
  - 3D Gaussian splatting as Digital Twin for the scene (Mill/Factory)
- AI constracted content:
  - **Object detection based on pre-trained classification model or Vision Language Model (VLM)**
  - Digital Twin instantiation based on ”external metadata” and visual ”facts”
    - **Pump**, color matches real-world pump, state read from real-time system
    - **Tank** level animated 0-100%, size matches to real tank
    - **Valve** opening read from real-time system and show status
  - **Deep links** created during instantiation, url parameters are found from metadata like **serial number** & **model**
    - Asset Management: documents & spare parts etc.
    - Condition Monitoring
  - Analytics and other AI based data from MCP Servers (actual presentation 10.12. OPC Day Finland, AWS Office)

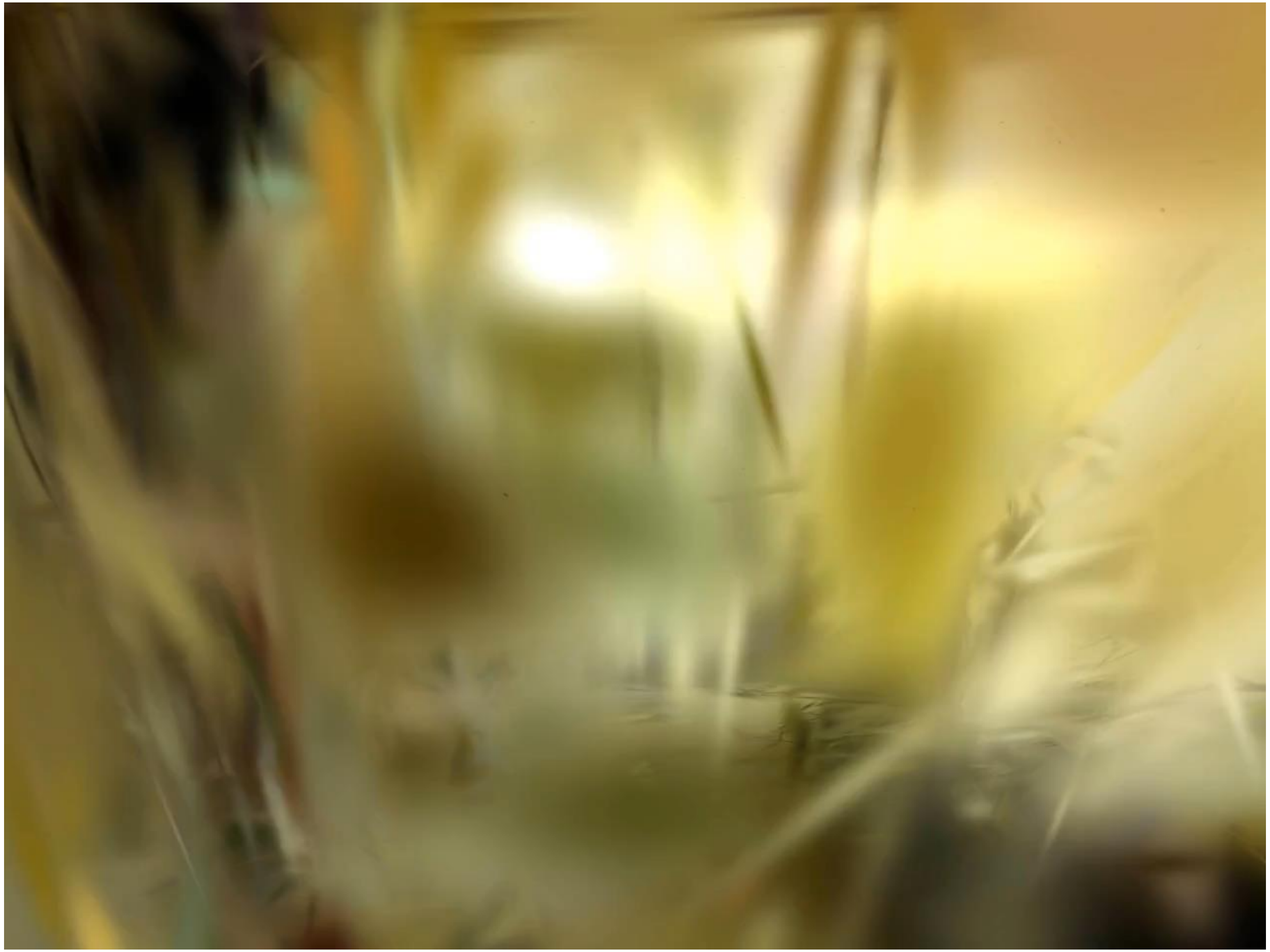
Comparison splat and mesh

3D Digital Twin based on Gaussian splatting



3DGS – Digital Twin from Valmet pilot plant by Nokia team

Constructed from short video & photos



Example: Device Maintenance Avatar

Guide safe route inside mill

Example: Real-time values and animated symbols

AI agents will insert & fit small Digital Twins during runtime to the scene, connecting to real-time values
Size & color and other features + deep links constructed by the AI agents

Summary – AI comes and makes things “automatically”

- Easier to build environment, virtual or 3D Gaussian splatting / mesh
- Maintains it, user will provide photos or short videos
- Most of the content is bind during the runtime & visualized (**like object detection**)

- Dynamic AI actions will need local on-premise or on-device AI model execution
  - Small Language Model (SML)
  - AI agents (can be run parallel)
    - **Object detection to find field devices, needed to insert device specific digital twins**
  - Good design: asynchronous & robust
  - Security is MUST
  - Context design instead of prompt design
  - Skills (methodology) will be solution

TIPS & Resources

- 3D Gaussian splatting (.ply) -> Mesh (online converters)
- Segment Anything (SAM v3 by Meta)

- Model available (AutoML pre-trained) for TensorFlow:
  - Contains Classifier

- 3D Gaussian splatting examples:
  - Pilot plant
  - Home automation
  - Library
  - Coffee place

Valmet
FORWARD